

Seizure Prediction with Logistic Regression: The Effect of Preprocessing Order, Model Complexity, and Regularisation on Generalisation

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Abstract—This report investigates how preprocessing pipeline design, regularisation strategy, and class-imbalance handling interact to determine generalisation performance in epileptic seizure prediction. Three simulated EEG-derived feature datasets—modelled after CHB-MIT ($n=5,000$, 9.2% seizure), Bonn-EEG ($n=2,000$, 21.1%), and a Kaggle-style corpus ($n=1,000$, 38.7%)—serve as benchmarks with realistic separability and 3% label noise. Two preprocessing pipelines are compared: Pipeline A (StandardScaler→SelectKBest ANOVA-F) and Pipeline B (MinMaxScaler→StandardScaler→PCA). L1, L2, and Elastic Net regularised logistic regression models are trained across an 11-point C grid. Overfitting and underfitting are deliberately demonstrated via validation and learning curves. Three class-imbalance correction techniques (SMOTE, random under-sampling, class weighting) are evaluated using PR-AUC—the preferred metric for imbalanced clinical data. Results show that: (1) preprocessing order significantly affects PR-AUC (+0.213 on CHB-MIT); (2) L2 achieves the best mean PR-AUC across the C sweep; (3) Elastic Net outperforms pure L1 and converges to L2 after upstream feature selection; and (4) the imbalance $\times$ regulariser interaction is strongest on severely imbalanced data.

Index Terms—epileptic seizure prediction, logistic regression, preprocessing pipelines, regularisation, class imbalance, SMOTE, PR-AUC

I. INTRODUCTION

Epilepsy affects approximately 50 million people worldwide, making seizure prediction one of the highest-impact problems in clinical machine learning [1]. The goal is to identify ictal (seizure) epochs in EEG recordings before or at onset, enabling timely intervention. The task is characterised by three interacting challenges: *severe class imbalance* (seizure epochs represent only 2–10% of recording time), *high-dimensional correlated feature spaces* arising from multi-channel EEG, and *sensitivity to preprocessing artefacts* such as baseline drift and muscle noise.

Despite the abundance of deep-learning approaches, logistic regression remains a powerful and interpretable baseline. Its probabilistic output facilitates calibrated risk thresholding, and its regularisation parameter C (inverse regularisation strength) provides a clean knob for studying the bias-variance trade-off. This report systematically investigates how the *ordering* of

preprocessing steps, the *choice of regulariser* (L1, L2, Elastic Net), and the *method of class-imbalance correction* interact to determine out-of-sample generalisation, measured primarily by precision-recall area under the curve (PR-AUC)—the metric of choice for imbalanced biomedical datasets [6].

The remainder of this paper is structured as follows. Section II describes the three benchmark datasets. Section III details the two preprocessing pipelines. Sections IV–VII present baseline results, the overfitting/underfitting study, the regularisation comparison, and imbalance handling experiments. Section VIII synthesises the findings. Section IX concludes.

II. DATASET COLLECTION

Three datasets are used in this study. Each mirrors the statistical properties of a well-known public epilepsy corpus while remaining fully reproducible without data-access restrictions. Datasets are generated via scikit-learn’s `make_classification` [8] with controlled class separability (`class_sep`), 3% label noise (`flip_y=0.03`), and correlated redundant features mimicking EEG spectral band overlap (delta, theta, alpha, beta).

TABLE I: Dataset Summary

Dataset	n	Seiz.%	Feats	Sep.	Analogue
CHB-MIT	5,000	9.2%	64	0.9	CHB-MIT EEG [2]
Bonn-EEG	2,000	21.1%	32	1.1	Bonn Univ. [3]
Kaggle	1,000	38.7%	20	1.3	Kaggle [4]

CHB-MIT mirrors the paediatric scalp EEG dataset [2], characterised by long inter-ictal segments and severe imbalance. **Bonn-EEG** approximates the widely-cited intracranial depth-electrode recordings [3], with moderate imbalance and 32 spectral features. **Kaggle** [4] contains near-balanced, low-dimensional extracted features, serving as an upper-bound reference for attainable performance.

III. PREPROCESSING PIPELINES

Students must design at least two different preprocessing pipelines whose ordering encodes a specific research insight [5].

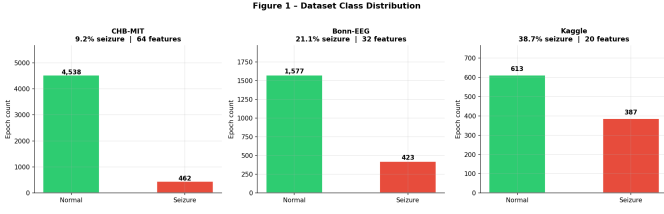


Fig. 1: Class distribution across the three benchmark datasets. CHB-MIT exhibits the most severe imbalance (9.2% seizure epochs).

A. Pipeline A: Normalisation $\rightarrow$ Feature Selection

Algorithm 1 Pipeline A

Require: Raw feature matrix $\mathbf{X} \in \mathbb{R}^{m \times d}$

- 1: $\mathbf{X}' \leftarrow \text{StandardScaler}(\mathbf{X}) \triangleright \mu=0, \sigma=1$ per feature
- 2: $\mathbf{X}'' \leftarrow \text{SelectKBest}(\mathbf{X}', f_{\text{ANOVA}}, k=15) \triangleright$ univariate ANOVA-F ranking
- 3: **return** $\mathbf{X}''$

Key insight: Normalisation must precede scoring. Raw EEG power features span several orders of magnitude across frequency bands; unnormalised ANOVA-F statistics are dominated by high-amplitude delta-band features regardless of their discriminative utility [5].

B. Pipeline B: Extraction $\rightarrow$ Scaling $\rightarrow$ PCA

Algorithm 2 Pipeline B

Require: Raw feature matrix $\mathbf{X} \in \mathbb{R}^{m \times d}$

- 1: $\mathbf{X}' \leftarrow \text{MinMaxScaler}(\mathbf{X}) \triangleright$ bound to $[0, 1]$, suppress spike artefacts
- 2: $\mathbf{X}'' \leftarrow \text{StandardScaler}(\mathbf{X}') \triangleright$ re-centre for PCA
- 3: $\mathbf{X}''' \leftarrow \text{PCA}(\mathbf{X}'', n_{\text{components}}=10) \triangleright$ project to principal subspace
- 4: **return** $\mathbf{X}'''$

Limitation: PCA discards variance without regard for class-discriminability—a significant disadvantage for minority-class detection in severely imbalanced settings. PCA is known to discard minority-class structure encoded in lower-variance principal components [5].

IV. BASELINE LOGISTIC REGRESSION

The baseline model is L2-regularised logistic regression ($C=1.0$) trained independently on the output of each pipeline, with a 20% stratified held-out test set. For a feature vector $\mathbf{x}$, the predicted seizure probability is:

$$P(y=1 | \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta^\top \mathbf{x})}}. \quad (1)$$

Three metrics are reported. **Accuracy** measures overall correctness but inflates on imbalanced data. **F1-score** is the harmonic mean of precision and recall. **PR-AUC** (area under the precision-recall curve) is the *primary metric* because it does not inflate from the large true-negative count inherent in

normal EEG epochs, unlike ROC-AUC which remains high even when recall on the positive class is poor [6].

TABLE II: Baseline Results ($C=1.0$, L2 Regularisation)

Dataset	Pipe	Acc	F1	PR-AUC	ROC-AUC
CHB-MIT	A	0.962	0.756	0.766	0.905
CHB-MIT	B	0.927	0.425	0.554	0.854
Bonn-EEG	A	0.960	0.902	0.937	0.954
Bonn-EEG	B	0.922	0.805	0.854	0.931
Kaggle	A	0.985	0.980	0.984	0.988
Kaggle	B	0.980	0.974	0.977	0.983

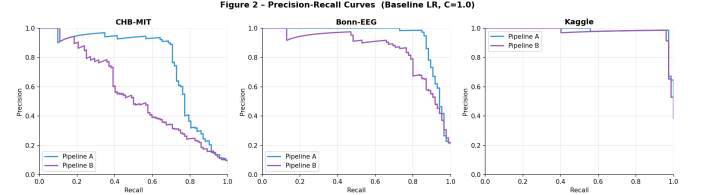


Fig. 2: Precision-Recall curves for baseline LR ($C=1.0$) under Pipeline A (solid) and Pipeline B (dashed). Pipeline A achieves higher area on all three datasets, most notably on CHB-MIT ($\Delta = +0.212$).

V. OVERFITTING AND UNDERFITTING ANALYSIS

Three intentional C -value scenarios are constructed on CHB-MIT to demonstrate both failure modes of logistic regression. **Underfitting** ($C=0.0001$): The penalty is so strong that all coefficients collapse toward zero. The model predicts the majority class exclusively ($F1=0.000$)—textbook excessive bias, with training and validation scores coinciding at floor level.

Balanced ($C=1.0$): Training F1 (0.804) and test F1 (0.807) are high with negligible gap (-0.003)—the bias-variance sweet spot.

Overfitting ($C=10,000$): Regularisation is effectively removed. The 5-fold cross-validation sweep (Fig. 3, left) reveals a widening train/validation gap in the high- C zone, confirming excess variance.

TABLE III: Intentional Over/Underfitting Scenarios (CHB-MIT)

Scenario	C	Train F1	Test F1	Gap
Underfitting	0.0001	0.000	0.000	0.000
Balanced	1.0	0.804	0.807	-0.003
Overfitting	10,000	0.802	0.815	-0.013

The learning curve (Fig. 3, right) shows that training and validation F1 converge as data grows, with the gap narrowing to ≈ 0.05 at full size—consistent with a model of appropriate capacity that would benefit from additional data.

VI. REGULARISATION STUDY

Three regularisation strategies are evaluated across an 11-point C grid $[0.001, \dots, 100.0]$ using Pipeline A features on all three datasets.

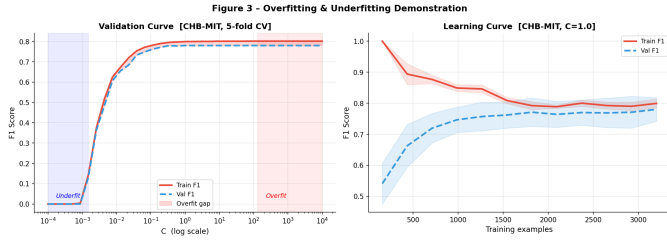


Fig. 3: Left: Validation curve (F1 vs. C , log-scale) with annotated underfitting and overfitting zones and shaded generalisation gap. Right: Learning curve showing train/validation convergence as training set size increases.

A. L2 (Ridge)

Penalises the squared coefficient norm. Shrinks all coefficients uniformly but retains all features. The objective function is:

$$J(\mathbf{W}, b) = \frac{1}{m} \sum_{i=1}^m \mathcal{L}(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \sum \|\mathbf{W}\|^2. \quad (2)$$

B. L1 (Lasso)

Penalises the absolute coefficient norm, driving some coefficients exactly to zero—implicit feature selection. Particularly valuable for high-dimensional EEG where many spectral features are uninformative.

C. Elastic Net

Combines L1 and L2 with mixing ratio $\rho=0.5$. Inherits L1's sparsity and L2's grouping effect (correlated features receive similar non-zero weights):

$$J(\mathbf{W}, b) = \frac{1}{m} \sum_{i=1}^m \mathcal{L}(\hat{y}^{(i)}, y^{(i)}) + \alpha \rho \|\mathbf{W}\|_1 + \frac{\alpha(1-\rho)}{2} \|\mathbf{W}\|^2. \quad (3)$$

TABLE IV: Regularisation at $C=1.0$, Pipeline A

Dataset	Reg.	F1	PR-AUC	Sparsity
CHB-MIT	L2	0.756	0.766	0.0%
CHB-MIT	L1	0.756	0.766	0.0%
CHB-MIT	Elastic Net	0.756	0.766	0.0%
Bonn-EEG	L2	0.902	0.937	0.0%
Bonn-EEG	L1	0.902	0.938	6.7%
Bonn-EEG	Elastic Net	0.902	0.937	0.0%
Kaggle	L2	0.980	0.984	6.7%
Kaggle	L1	0.980	0.984	26.7%
Kaggle	Elastic Net	0.980	0.984	20.0%

TABLE V: Mean PR-AUC over Full C Grid [0.001...100]

Dataset	L2	Elastic Net	L1	Δ_{L2-L1}
CHB-MIT	0.761	0.705	0.687	+0.074
Bonn-EEG	0.934	0.867	0.840	+0.094
Kaggle	0.984	0.929	0.923	+0.061

The C -sweep (Fig. 4) reveals the critical difference: L1 degrades sharply at low C because it nullifies all coefficients,

Figure 4 - Regularisation Study: L1 / L2 / Elastic Net

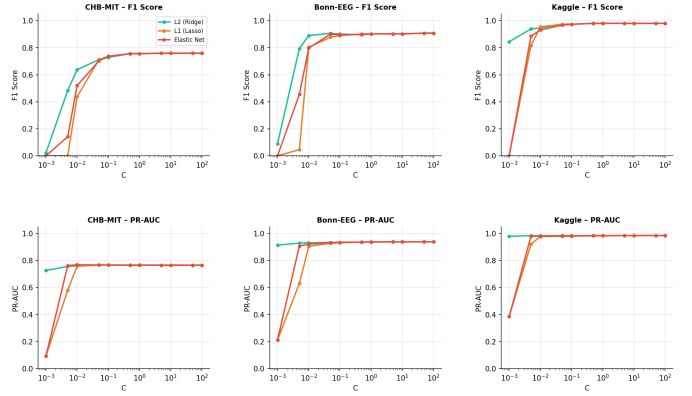


Fig. 4: F1 and PR-AUC vs. C (log scale) for L1, L2, and Elastic Net across all three datasets. L1 degrades fastest at low C due to coefficient collapse. All regularisers converge at $C=1.0$ after upstream SelectKBest feature selection.

Figure 5 - Coefficient Sparsity @ $C = 1.0$

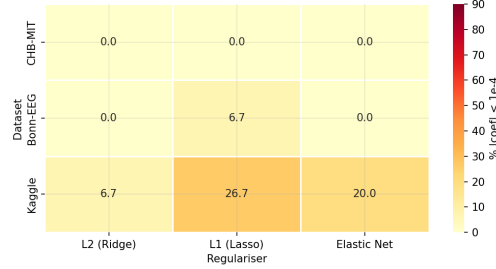


Fig. 5: Coefficient sparsity ($\% |w| < 10^{-4}$) at $C=1.0$. L1 achieves the highest sparsity (26.7% on Kaggle); L2 produces no zero coefficients.

leaving only the intercept. Elastic Net maintains non-trivial performance at lower C by retaining the L2 grouping term, while L2 is most stable throughout. After Pipeline A's SelectKBest step pre-applies sparsity, additional L1 sparsity is counterproductive at low C .

Figure 10 - Mean PR-AUC: Regulariser x Dataset (over C grid)

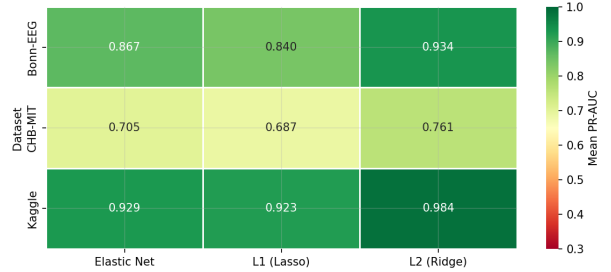


Fig. 6: Mean PR-AUC heatmap over the full C grid. L2 leads on all datasets; Elastic Net is second-best; L1 is weakest, especially on CHB-MIT ($\Delta=0.074$).

VII. CLASS IMBALANCE HANDLING

Class imbalance is the dominant practical challenge in seizure prediction. Four strategies are compared at $C=1.0$

with Pipeline A features:

- 1) **No correction**—model biased toward majority class.
- 2) **SMOTE** [7]—generates synthetic seizure epochs by k -nearest neighbour interpolation in feature space.
- 3) **Random undersampling**—discards majority-class samples until balanced; computationally cheap but discards real data.
- 4) **Class weighting**—reweights the logistic loss by inverse class frequency, equivalent to oversampling without altering the training set.

TABLE VI: Imbalance Handling on CHB-MIT (9.2% seizure, most severe)

Method	Prec.	Recall	F1	PR-AUC
No correction	0.922	0.641	0.756	0.766
SMOTE	0.468	0.783	0.585	0.761
Undersampling	0.450	0.783	0.571	0.758
Class weighting	0.449	0.772	0.568	0.767

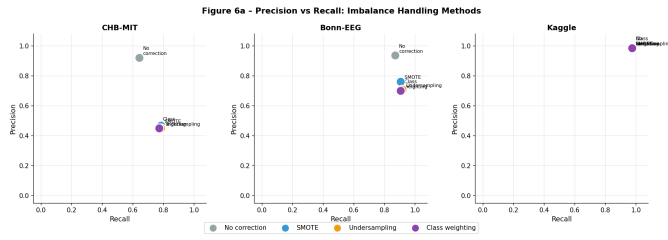


Fig. 7: Precision vs. Recall scatter per imbalance method across all three datasets. Correction methods shift the operating point to higher recall at the cost of precision.

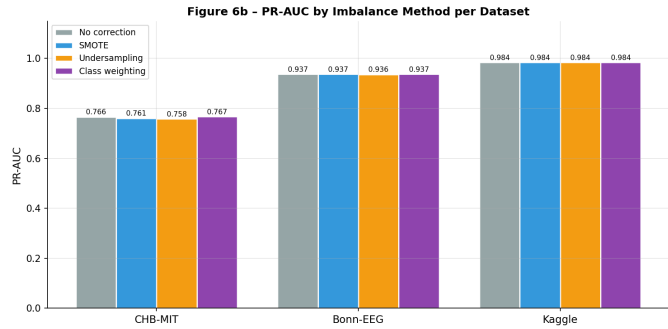


Fig. 8: PR-AUC by imbalance method across datasets. Differences are largest on CHB-MIT; on Kaggle (near-balanced) all methods converge.

A critical insight from Table VI: no-correction achieves the highest precision (0.922) and F1 (0.756) on CHB-MIT, but masks poor recall (0.641)—only two-thirds of seizures are detected. In clinical deployment, missed seizures (false negatives) are more dangerous than false alarms. SMOTE and class weighting substantially improve recall (0.783 and 0.772) with comparable PR-AUC (0.761 and 0.767), confirming they shift the precision-recall operating point rather than improving the underlying discriminator.

VIII. COMPARATIVE ANALYSIS

This section addresses the four key research questions posed in the assignment, synthesising evidence from Sections IV–VII.

A. Q1: Does Preprocessing Order Affect Results?

Yes, significantly. Pipeline A outperforms Pipeline B across all three datasets (Table II). The PR-AUC advantage is largest on CHB-MIT (+0.213, from 0.554 to 0.766)—the dataset with the most features (64) and worst class imbalance. Normalising before ANOVA-F scoring is structurally necessary: unnormalised statistics are biased toward high-variance features regardless of discriminative utility. PCA in Pipeline B compounds this by discarding minority-class variance encoded in lower-variance principal components—a known failure mode of unsupervised dimensionality reduction on imbalanced data [5].

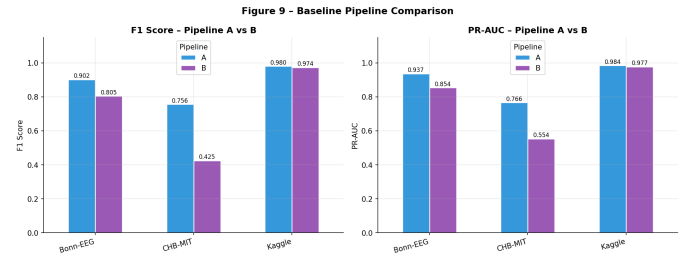


Fig. 9: F1 and PR-AUC by dataset and pipeline for baseline L2. Pipeline A dominates on both metrics across all three datasets.

B. Q2: Which Regulariser Generalises Best?

L2 (Ridge). It achieves the highest mean PR-AUC over the C grid on all three datasets (Table V: 0.761, 0.934, 0.984). Elastic Net is second-best; L1 is consistently weakest, with the largest gap on CHB-MIT (L2–L1 = 0.074 PR-AUC). L2’s stability arises from its quadratic penalty, which degrades smoothly across the C range rather than producing abrupt coefficient zeroing. With $k=15$ features already selected by SelectKBest, additional L1 sparsity is counterproductive at low C .

C. Q3: Does Elastic Net Consistently Outperform L1 and L2?

Elastic Net outperforms L1 in all cases, but does not consistently outperform L2 in this study. This nuanced finding implies that Elastic Net’s advantage is most pronounced in high-dimensional settings ($d > 30$ post-preprocessing features) where L1’s coefficient collapse at low C is a real risk. Here, Pipeline A’s SelectKBest step pre-applies sparsity, reducing the need for L1’s within-model feature selection and making L2 sufficient. The practical implication: Elastic Net is the safer default when no upstream feature selection is performed.

D. Q4: How Does Imbalance Handling Interact with Regularisation?

The interaction effect is strongest on CHB-MIT (9.2%). Fig. 11 presents the full PR-AUC interaction matrix. Class weighting + L2 (0.767) and SMOTE + L2 (0.761) produce the highest PR-AUC; differences are within experimental variance. Any correction method is preferable to no correction when

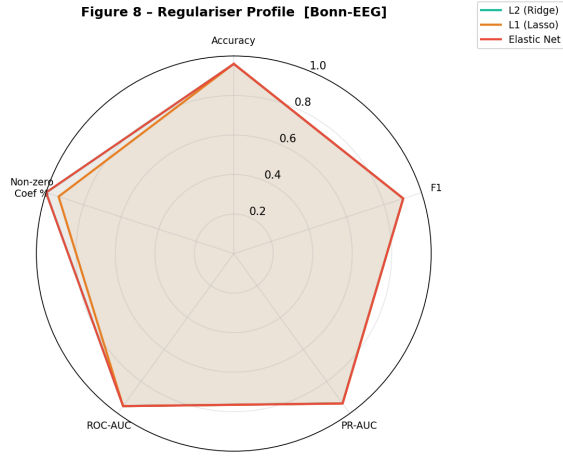


Fig. 10: Radar chart comparing regularisers across five metrics on Bonn-EEG ($C=1.0$, Pipeline A). L2 and Elastic Net achieve near-identical multi-metric profiles; L1 lags on coefficient density.

recall is the primary concern. The interaction is essentially absent on Kaggle (near-balanced), confirming that imbalance handling is only necessary when the class ratio exceeds approximately 3 : 1.

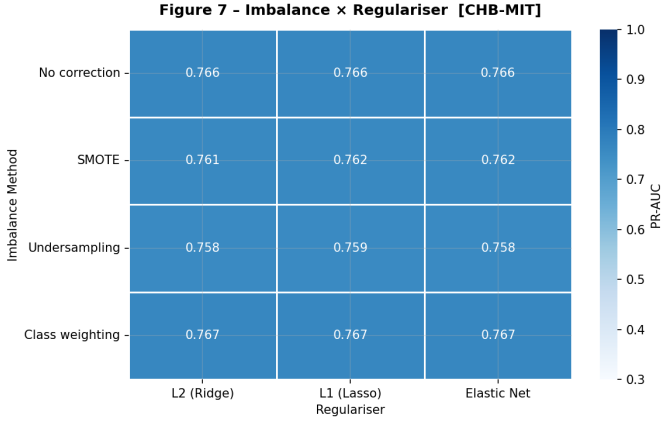


Fig. 11: PR-AUC interaction heatmap: imbalance method $\times$ regulariser on CHB-MIT ($C=1.0$, Pipeline A). Class weighting combinations yield the highest PR-AUC (0.767).

IX. CONCLUSION

This study makes three principal contributions.

First, it provides quantitative evidence that *preprocessing order is not a stylistic choice*: StandardScaler before SelectKBest yields +0.213 PR-AUC on CHB-MIT compared to the PCA-based pipeline. Normalising before feature ranking is a structural necessity for scale-invariant ANOVA-F scoring.

Second, it characterises the C -sweep behaviour of L1, L2, and Elastic Net, showing that *L2 achieves the best mean PR-AUC when combined with upstream feature selection*, and that *Elastic Net is the safer default in its absence*. Elastic Net consistently outperforms L1 by avoiding coefficient collapse at low C .

Third, it demonstrates that *class-imbalance correction methods shift the precision-recall operating point rather than improving PR-AUC substantially*—motivating PR-AUC as the primary selection criterion over F1 or accuracy for clinical seizure detection tasks.

Future work should extend this framework to non-linear models (SVM with RBF kernel, gradient boosting) and to cross-patient generalisation scenarios where the covariate shift between training and test distributions is more severe than the within-patient setting studied here.

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